UNIVERSITY
OF COLOGNE

WORKSHOP 01

Advanced Analytics and Applications

Agenda

0	**Course Admin**
1	Lecture Recap
2	Exemplary Exam Questions
3	Coding Tasks

What is the purpose of these workshops?

- First, the workshop will usually **recap** the most important concepts from lecture
- Then we will discuss some **exemplary exam-type questions** and solutions
- Finally, we will go over a **Jupyter Notebook** that gives you the most important tools for carrying out the learned concepts yourself and complete the team assignment.
- The Jupyter Notebook we hand out weekly is best suited for **self-study**!
  We will go over the most important stuff but not the whole thing line by line.
  If questions come up please ask them in the next workshop.
  The pre-workshop version has some tasks without solution code, we will talk about this in the workshop but you can try to prepare it before.
- We will try to focus on how to **translate concepts into applications**.

Python: The programming language of choice

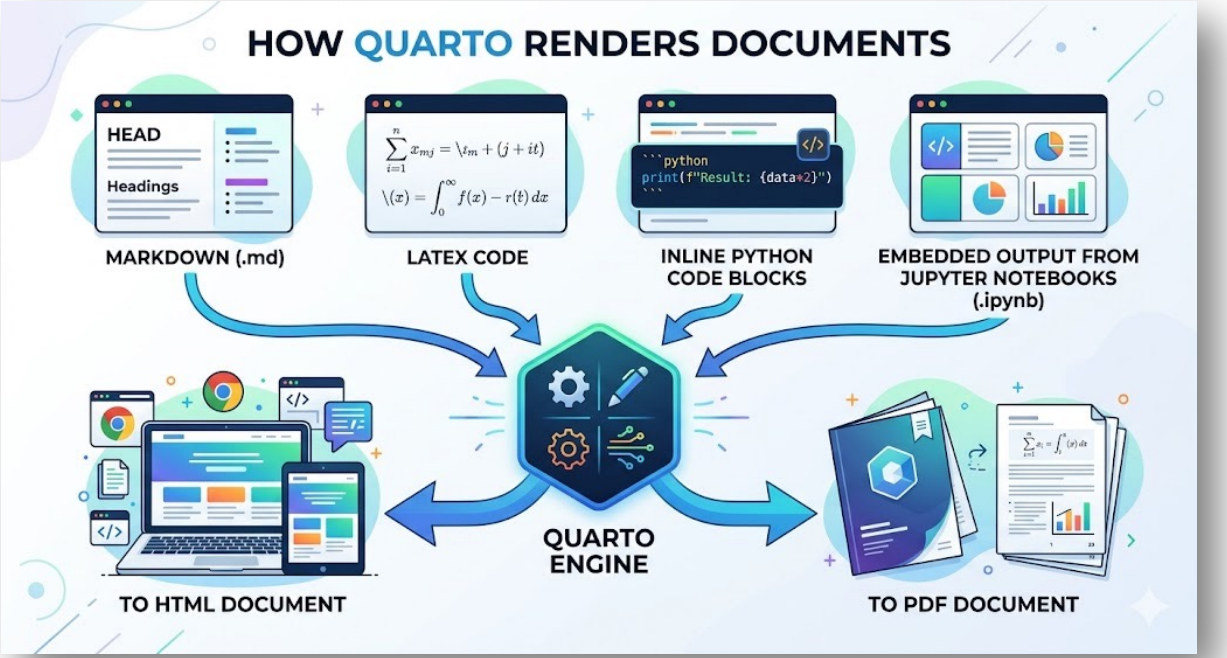
- Python is fully open source, including all packages that extend it.
- Other languages are more specialized for statistics (e.g. R) or speed (e.g. Julia), but Python is a good general purpose language, which is used for a lot of things and is thus well established.
- It is highly extensible and the community developing packages for it is vast.
- You can find almost every answer you might have on sites such as Stack Overflow
- Because of it's popularity, Python is pretty well established with LLM.
  (ChatGPT and the likes of it can understand and help with your code, but be careful!)

Team Assignment

- **Sign up** for the combined exam by the end of next week (April 27th, 23:59)!
- **No participation without registration to the exam!**
- Teams will be **allocated randomly**, between 3–6 students per team.
- The submission needs to be a public **GitHub repository** (or alternative, like gitlab)
- Team assignments start in two weeks (April 29)

Team Assignment Submission

- Starting with this year, there is a mandatory assignment template you must follow for comparability
- We use **Quarto Reports**, which are dynamic documents
- This allows **full transparency into the collaborative writing process**, as changes to the report must also be committed to your repository and we can analyze the commit history
- No participation (no commits to either code or report text) –> No passing grade
- We reserve the right to additionally use this information for individual grades within a team



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Lecture Slide

Assessment Guidelines

50%

Written Exam

- **Scope:** 90 minutes, closed book
- **Content:** All materials provided in the Lectures, Tutorials, and Assignments.
- **Format:** Written and multiple choice questions
- **Dates:**
  - **Exam 1: August 10**
  - **Exam 2: tba**

50%

Team Assignment

- **Set-up**: Teams of 3-6 students
- **Tasks**:
  - Data Science Project,
    from Business Problem through Deployment
  - Reinforcement Learning Toy Problem
- **Deliverables**:
  Github Repository with Report
- **Deadline: July 26**

Machine Learning Basics

- As this is an **advanced** course, we expect students to either have taken one of our prior courses (such as *Introduction to Data Science and Machine Learning* or *Analytics and Applications*) **or** read up on concepts themselves!
- If you did not take a prior basic machine learning course, it will be hard to follow along for some topics (e.g., “advanced classification”, “advanced clustering” 😉)
- We recommend to go through the relevant concepts using the materials from *Analytics and Applications* which you can access at this ILIAS folder: https://www.ilias.uni-koeln.de/ilias/goto.php/crs/6460102 (previous semester, Password: AA_2025!)
- You are free to use other resources to read up on concepts, but please be aware that we cannot answer questions that arise from that. Every resource uses slightly different notation, and we make sure that notation stays the same within our courses.

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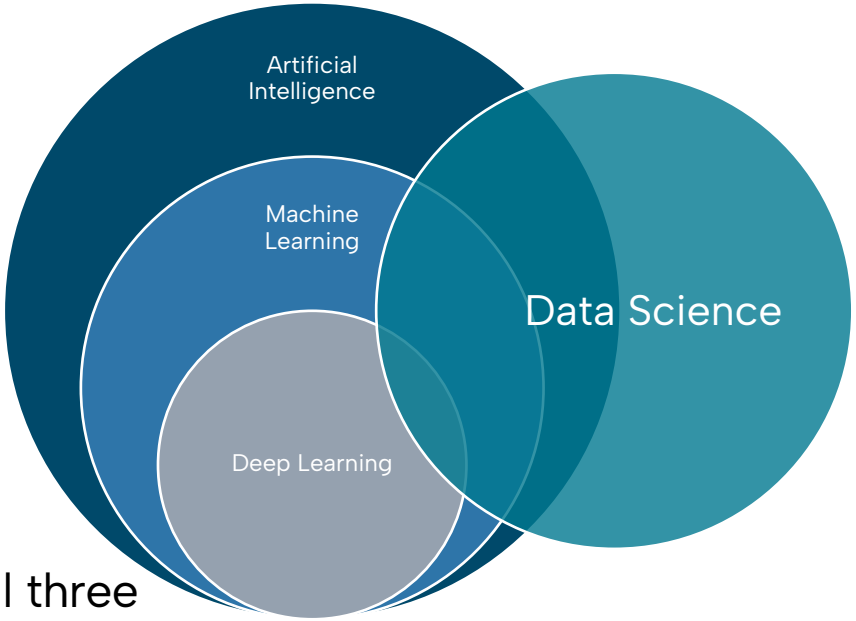
“Data Science” vs “Machine Learning” vs “AI”

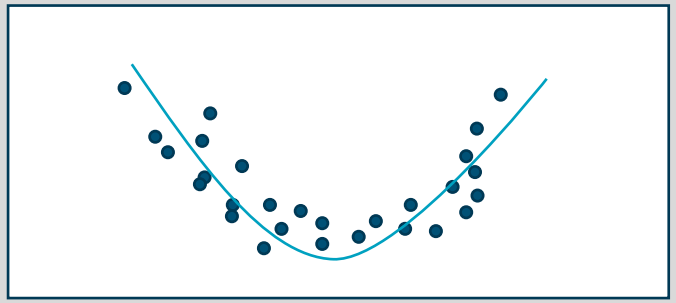
- We differentiate the different concepts based on the scope and the methods used:
  - **Artificial Intelligence** is the field of **mimicking human behavior** / intelligence with machines (can be as simple as rule-based)
  - **Machine Learning** is the field of **creating (predictive) models** for a task that performs well with regard to some performance measure based on experience (data) [definition by Mitchell]
  - **Deep Learning** is the subfield of ML that utilizes **deep neural networks** as the model. Here, we trade model explainability for hard-to-argue performance (in large tasks).
  - **Data Science** is the field of **using data to derive insights and decisions** for a business / application context and overlaps all three



Lecture Slide

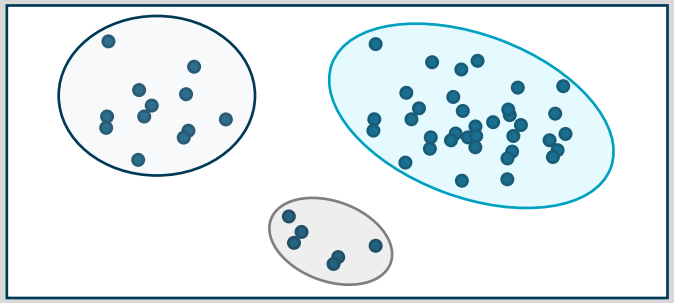
Three core Machine Learning concepts

Supervised Learning



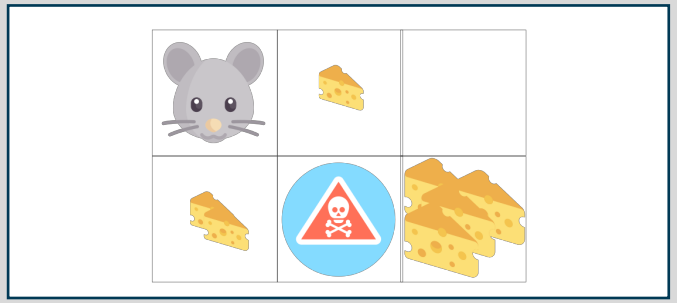
- Availability of **labeled** data
- Goal to learn a model that describes the relationship of **input features** and label
- Differentiation between **regression** (i.e. typically continuous targets) and **classification**
- Model performance **relatively easy** to evaluate

Unsupervised Learning



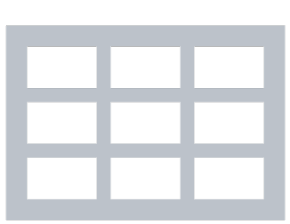
- Data **without** labels
- Goal to find certain structural **patterns** within the data
- Find **clusters** in data with similar characteristics
- Model performance **hard** to evaluate

Reinforcement Learning



- Sequence of **coupled decisions** instead of one-shot predictions
- Goal to improve **dynamic decision processes** by reacting and adapting to **reward signals**
- Connection to **game theory**
- Building block of **artificial intelligence** that strongly resembles human decision making

Machine Learning Models (Supervised)

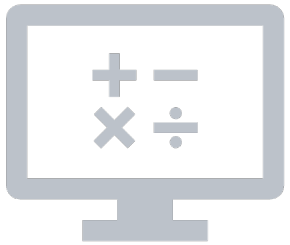


Data

- Observations
- Features X
- Target variable y

Preprocessing

- Feature Selection $X^* \subset X$
- Outlier Detection



Training

- Minimization of Objective $\min \sum l(y - y')$
- Loss based on target y, predicted $\hat{y}$ and error metric l

Model

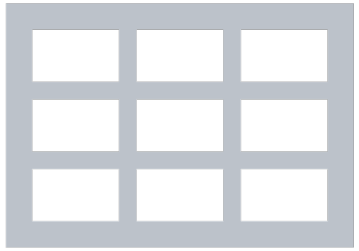
- Statistical model consisting of learned parameters θ
- Used to predict outcomes $\hat{y}$ from new

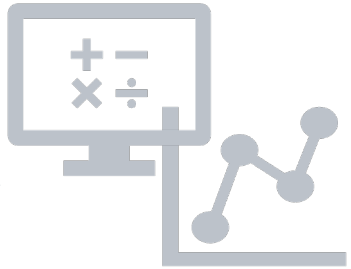
Feature Types: Discrete, Numerical
Target Variable Types:

- Discrete > **Classification**
- Numerical > **Regression**

Metrics: R^2, MSE, Accuracy, etc.

Machine Learning Models (Unsupervised)





Data
- Observations
- Features X

Preprocessing
- Feature Selection $X^* \subset X$
- Outlier Detection
- Sampling / Filtering

Finding a Model (Representation)
- Minimization of Objective $\min \sum l(X, X')$
- Loss based comparison of observations X, X' and (dis)similarity metric l

Metrics: Cosine Similarity, Euclidean Distance, etc.

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Multiple Choice Questions: Machine Learning

Question: Astronomers have been cataloguing distant objects in the sky using long-exposure CCD images. The objects need to be labeled as star, galaxy, nebula etc. The data is highly noisy, and the images are very faint. The cataloguing can take decades to complete. Which Method would you recommend the physicists use to automate the cataloguing process, and improve its effectiveness?

- ❑ **Regression** (Supervised)
- ❑ **Classification** (Supervised)
- ❑ **Clustering** (Unsupervised)

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Multiple Choice Questions: Machine Learning

Question: A customer of a consultancy owns a supermarket. Through membership cards, there is basic data about the supermarket customers like Customer ID, age, gender, annual income and spending score. Spending Score was assigned to the customers based on defined parameters like customer behavior and purchasing data. Which method would you recommend using so the consultancy can make a suggestion to the supermarket which of his customers are most likely to react positively to a marketing campaign?

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Multiple Choice Questions: Machine Learning

Question: CComp, a large company that employs around 4000 employees, and which faces a significant annual employee attrition rate, seeks to understand the underlying factors driving this turnover. To mitigate this issue, they want to identify actionable workplace changes that will encourage employee retention. Which analytical method can best model the probability of attrition and highlight the most impactful variables for the management to address?

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Multiple Choice Questions: Machine Learning

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- ☐ **Regression** (Supervised)
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- ☐ **Clustering** (Unsupervised)

Calculation: Error Metrics

a) Write down the Mean Squared Error (MSE) formula:

b) Calculate MSE from the sample below:

x	y	$\hat{y}$	$(y - \hat{y})^2$
1	2	2.8	
2	4	3.4	
3	5	4	
4	4	4.6	
5	5	5.2	

Calculation: Error Metrics

a) Write down the Mean Squared Error (MSE) formula:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

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$$MSE = \frac{1}{5}\sum_{i=1}^{5}(y_i - \hat{y}_i)^2 = \frac{1}{5} * \cdots$$

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2	4	3.4	0.36
3	5	4	1.00
4	4	4.6	0.36
5	5	5.2	0.04

$$MSE = \frac{1}{5}\sum_{i=1}^{5}(y_i - \hat{y}_i)^2 = \frac{1}{5} * 2.4 = \frac{2.4}{5} = 0.48$$

Calculation: Error Metrics

c) Explain the R^2 formula by its components.

d) Calculate R^2 from the sample below:

x	y	$\hat{y}$	$(y-\hat{y})^2$	$(y-\bar{y})^2$
1	2	2.8	0.64	
2	4	3.4	0.36	
3	5	4	1.00	
4	4	4.6	0.36	
5	5	5.2	0.04	

Calculation: Error Metrics

c) Explain the R^2 formula by its components.

$$R^2 = 1 - \frac{SS\boldsymbol{E}}{SS\boldsymbol{T}} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Alternative representation with sum of squares of the **R**egression line:

$$R^2 = \frac{SSR}{SST} = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

The R^2 is a proportion of how much of the **T**otal sum of squares ($\sum_{i=1}^{n}(y_i - \bar{y})^2$) cannot ($1 - \cdots$) be captured with the regression, by looking at the residual **E**rrors ($\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$).

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3	5	4	1.00	
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5	5	5.2	0.04	

$$\bar{y} = \frac{2 + 4 + 5 + 4 + 5}{5} = 4$$

Calculation: Error Metrics

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d) Calculate R^2 from the sample below:

x	y	$\hat{y}$	$(y - \hat{y})^2$	$(y - \bar{y})^2$
1	2	2.8	0.64	4
2	4	3.4	0.36	0
3	5	4	1.00	1
4	4	4.6	0.36	0
5	5	5.2	0.04	1

$$\bar{y} = \frac{2 + 4 + 5 + 4 + 5}{5} = 4$$

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{2.4}{6} = 0.6$$

Calculation: Error Metrics

e) When is the MSE a useful metric?

The MSE puts larger emphasis on large deviations from the true value, which makes it useful when large errors have more significance than small errors and we critically want to prevent them.

f) When is the R^2 a useful metric? Compare it to the MSE!

The R^2 presents a measure of how much variance of y can be modeled with X.
This is useful if we want to compare different models (e.g., with a different feature set) on the same training data set.

The R^2 is unitless, making it more robust to scales of the features but can still be affected by outliers.

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Contact

For general questions and enquiries on **research**, **teaching**, **job openings** and **new projects** refer to our website at www.is3.uni-koeln.de

For specific **enquiries regarding this course** contact us by sending an email to the **IS3 teaching address** at is3-teaching@wiso.uni-koeln.de

To help us process your request efficiently, please use the following subject line format:

[AAA] <Your request subject>